Feiyang Liu

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Research Interests

My research aims to understand how multimodal foundation models learn: how visual and linguistic representations emerge and interact during training, and which properties of a training recipe determine its outcome.

- **Training mechanisms:** how understanding, generation, and language interact in unified models, and how data mixtures, loss weighting, and learning rates shape what they learn (UMP).
- **Probing and evaluation:** probes of internal representations, also used as cheap evaluations (UMP).
- **LLMs for vision:** enabling LLMs to understand and generate visual content (PixelBench, GroundingBench).

Education

South China University of Technology Sep 2022 – Jun 2027 (expected)
B.Eng. in Intelligent Science and Technology, School of Automation Science and Engineering

University of California, Berkeley Aug 2024 – May 2025
Visiting Student

Publications

1. Haoquan Zhang, **Feiyang Liu**, Juanxi Tian, Wanhua Li, Haodong Duan, Weiyang Liu. **PixelBench: Can LLMs Understand and Generate Images Pixel by Pixel?** 2026. [Project]

2. Qinzheng Zhou, **Feiyang Liu**, Lijing Lu, Zhihang Li. **Confidence-Guided Multi-Scale Aggregation for Sparse-View High-Resolution 3D Gaussian Splatting.** *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 **(Highlight)** [Paper]

Research Experience

UMP: Unified Multimodal Models
Project Lead, Research Intern (remote); with UC Santa Cruz and Google DeepMind (Gemini)

- **Training:** leading full-pipeline training of unified multimodal models for image understanding and text-to-image generation on Qwen3 backbones, from large-scale pretraining to multi-task post-training, in JAX/Flax.
- **Infrastructure:** multi-host, multi-slice training on hundreds of Cloud TPU v5p/v6e chips; throughput optimization (sharding, ZeRO-1, splash attention, fused gradient sync, dynamic-length batching); automatic slice-loss fallback and a shared compilation cache.
- **Evaluation:** tracks text, understanding, and generation together (decontaminated GenEval, image-shuffle tests of image use); fast recipe validation by branch-and-decay annealing and short-SFT probes, with seed-noise floors and preset thresholds; testing whether cheap probes rank checkpoints like full evaluations.
- **Data:** multi-source mixtures for understanding, generation, and text streams, deduplicated and decontaminated.
- **Analysis:** ablations over data mixtures, loss weighting, learning rates, image preprocessing, training budget, and module decoupling; weight-interpolation probes; located where language ability is lost and diagnosed generation overfitting.

PixelBench and GroundingBench
Research Collaborator (remote); with The Chinese University of Hong Kong and ByteDance Seed

- *PixelBench*: can LLMs generate pixel images directly with their existing vocabulary? Found that pixel understanding tracks pixel generation, and that training models to reconstruct pixels helps visual understanding. Built the initial demo, proposed ideas, and designed part of the experiments.
- *GroundingBench*: contributing to a benchmark for visual grounding, including the task survey and the design of detection and counting tasks.

Sparse-View 3D Gaussian Splatting
Co-author, with collaborators from HUST, Peking University and Alibaba

- Studied how input resolution affects sparse-view 3DGS and proposed confidence-guided multi-scale aggregation (CVPR 2026); contributed to algorithm design and paper writing.

Skills

- **Programming & Graphics:** Python, C/C++, Java, GLSL, RISC-V assembly; OpenGL, Unity, 3DGS
- **Large-Scale Training:** JAX/Flax, PyTorch; multi-host, multi-slice Cloud TPU v5p/v6e training, FSDP-style sharding, ZeRO-1, XLA profiling, MFU analysis
- **Multimodal Models:** unified understanding-and-generation training, visual tokenizers, data mixing, evaluation
- **Outreach:** Tech content creator on Xiaohongshu (RED), sharing AI research discussion; 4,200+ followers